Computer Basics

A computer is an electronic device that accepts data (input), processes it according to instructions (programs), stores it, and produces results (output). Charles Babbage is known as the **Father of the Computer** for designing the Analytical Engine in the 1830s.

1. Main Parts of a Computer

- **CPU (Central Processing Unit):** the "brain" of the computer. It has the ALU (Arithmetic Logic Unit), which does calculations, and the CU (Control Unit), which directs operations.
- **Memory:** RAM is temporary (volatile) memory that loses data when power is off. ROM is permanent (non-volatile) memory that stores start-up instructions.
- **Storage:** HDD (hard disk drive), SSD (solid state drive, faster), and pen drives keep data permanently.
- **Motherboard:** the main circuit board that connects all components.

2. Input and Output Devices

Input Devices	Output Devices
Keyboard, Mouse	Monitor
Scanner, Microphone	Printer, Speakers
Webcam, Joystick	Projector, Headphones

3. Hardware and Software

Hardware is the physical part of a computer that we can touch. **Software** is a set of programs that tells the hardware what to do. Software is of two types:

- **System software:** runs the computer, e.g. Operating Systems such as Windows, Linux, macOS and Android.
- **Application software:** does specific tasks, e.g. MS Word, web browsers, games, and calculators.

4. Units of Memory

The smallest unit is a **bit** (0 or 1). 8 bits = 1 Byte; 1024 Bytes = 1 Kilobyte (KB); 1024 KB = 1 Megabyte (MB); 1024 MB = 1 Gigabyte (GB); 1024 GB = 1 Terabyte (TB).

5. Generations of Computers

- **First (1940–56):** vacuum tubes
- **Second (1956–63):** transistors
- **Third (1964–71):** integrated circuits (ICs)
- **Fourth (1971–present):** microprocessors
- **Fifth (present and beyond):** artificial intelligence (AI)

6. Important Terms

- **Internet:** a global network connecting millions of computers.
- **Virus:** a harmful program that can damage data; antivirus software protects against it.
- **Binary system:** computers understand only two digits, 0 and 1.